

## iCONNECT - 2026 PROGRAM SCHEDULE Venue: Premier University, Wasa Campus

### DAY 1: 25 SEPTEMBER 2026 • FRIDAY

| TIME                | PROGRAM DETAILS                                                                                                                             |                                                                                                     |                                                    |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|----------------------------------------------------|
| 08:30 AM – 09:00 AM | Opening Registration Booth                                                                                                                  |                                                                                                     |                                                    |
| 09:00 AM – 10:30 AM | <b>Parallel Technical Sessions:</b><br>PS1- Room 605: AIML-PS1<br>PS2- Room 606: AI/ML PS2<br>PS3- Room 607: AIML-PS3<br>PS4- Room 608: CES |                                                                                                     |                                                    |
| 10:30 AM – 11:30 AM | Inauguration at Auditorium                                                                                                                  |                                                                                                     |                                                    |
| 11:30 AM – 11:45 AM | Refreshment & Tea Break                                                                                                                     |                                                                                                     |                                                    |
| 11:45 AM – 12:45 PM | Keynote Session-I: <b>Dr. Khondkar Siddique-e-Rabbani</b><br>Keynote Session-II: <b>Dr. Celia Shahnaz</b>                                   |                                                                                                     |                                                    |
| 12:45 PM – 02:30 PM | Prayer & Lunch Break                                                                                                                        |                                                                                                     |                                                    |
| 02:30 PM – 04:00 PM | Invited Talk-1                                                                                                                              | <b>Parallel Technical Sessions:</b>                                                                 |                                                    |
|                     |                                                                                                                                             | PS5- Room 605: AIML-PS4<br>PS6- Room 606: AIML-PS5<br>PS7- Room 607: BME-HT<br>PS8- Room 608: PE-ES | VS1- AI/ML VS1<br>VS2- AI/ML VS2<br>VS3- AI/ML VS3 |
| 04:00 PM – 04:30 PM | Tea Break                                                                                                                                   |                                                                                                     |                                                    |
| 04:30 PM – 06:00 PM | Invited Talk-2                                                                                                                              | <b>Parallel Technical Sessions:</b>                                                                 |                                                    |
|                     |                                                                                                                                             | PS9- Room 605: CV-PR-NLP<br>PS10- Room 606: CS-SDE<br>PS11- Room 607: RE-SPT<br>PS12- Room 608: RIA | VS4- AIML-IA<br>VS5- PE-RE<br>VS6- CS-DE-QC        |
| End of Day 1        |                                                                                                                                             |                                                                                                     |                                                    |

## DAY 2: 26 SEPTEMBER 2026 • SATURDAY

| TIME                | PROGRAM DETAILS                                                                                                                                                                                                                                    |                                                                   |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| o8:30 AM            | Opening Registration Booth                                                                                                                                                                                                                         |                                                                   |
| 09:00 AM – 10:30 AM | <b>Parallel Technical Sessions:</b>                                                                                                                                                                                                                |                                                                   |
|                     | PS13- Room 605: AIML-PS6<br>PS14- Room 606: CSA<br>PS15- Room 607: PE-MD<br>PS16- Room 608: AIML                                                                                                                                                   | VS7- ST-ICA<br>VS8- IoT-IES<br>VS9- ES-RCA                        |
|                     | 10:30 AM – 11:00 AM                                                                                                                                                                                                                                | <b>Conference Tour: Parki Beach<br/>Through Karnaphuli Tunnel</b> |
|                     | 11:00 AM – 11:30 AM                                                                                                                                                                                                                                |                                                                   |
| 11:15 AM – 01:30 PM |                                                                                                                                                                                                                                                    |                                                                   |
| 01:30 PM – 02:30 PM |                                                                                                                                                                                                                                                    |                                                                   |
| 02:30 PM – 03:30 PM | Industry-Academia Collaboration Session in Association With YP BDS                                                                                                                                                                                 |                                                                   |
| 03:30 PM – 04:00 PM | <b>Invited Talk-3: Prof. Dr. Chandima Pathirana</b><br><b>Zoom Link:</b><br><b><a href="https://bdren.zoom.us/j/91764784693?pwd=5YPE1psQZQZ7TCdxjce9OQKHjMb8NL.1">https://bdren.zoom.us/j/91764784693?pwd=5YPE1psQZQZ7TCdxjce9OQKHjMb8NL.1</a></b> |                                                                   |
| 04:00 PM – 04:30 PM | Tea Break                                                                                                                                                                                                                                          |                                                                   |
| 04:30 PM – 05:00 PM | <b>Keynote Session-III: Dr. Sivaperumal Sampath</b>                                                                                                                                                                                                |                                                                   |
| 05:00 PM – 06:30 PM | Closing & Prize Giving Ceremony                                                                                                                                                                                                                    |                                                                   |
| 07:00 PM – 10:00 PM | Gala Dinner                                                                                                                                                                                                                                        |                                                                   |

## TECHNICAL PROGRAM SCHEDULE – AT A GLANCE

### DAY 1: 25 SEPTEMBER 2026 • FRIDAY- PHYSICAL (ON-SITE) • MORNING

| ID  | SHORT NAME | TIME                               | ROOM     | PAPER IDs                    |
|-----|------------|------------------------------------|----------|------------------------------|
| PS1 | AIML-PS1   | 25th Sep FRI<br>09:00 AM -10:30 AM | Room 605 | 32, 261, 269, 362, 544       |
| PS2 | AIML-PS2   | 25th Sep FRI<br>09:00 AM -10:30 AM | Room 606 | 377, 383, 421, 429, 590      |
| PS3 | AIML-PS3   | 25th Sep FRI<br>09:00 AM -10:30 AM | Room 607 | 598, 611, 627, 668, 674      |
| PS4 | CES        | 25th Sep FRI<br>09:00 AM -10:30 AM | Room 608 | 273, 373, 594, 633, 776, 911 |

| DAY 1: 25 SEPTEMBER 2026 • FRIDAY - PHYSICAL (ON-SITE) • AFTERNOON |            |                                    |          |                              |
|--------------------------------------------------------------------|------------|------------------------------------|----------|------------------------------|
| ID                                                                 | SHORT NAME | TIME                               | ROOM     | PAPER IDs                    |
| PS5                                                                | AIML-PS4   | 25th Sep FRI<br>2:30 PM- 04:00 PM  | Room 605 | 263, 704, 707, 710, 719,748  |
| PS6                                                                | AIML-PS5   | 25th Sep FRI<br>2:30 PM- 04:00 PM  | Room 606 | 777, 784, 805, 847, 896      |
| PS7                                                                | BME-HT     | 25th Sep FRI<br>2:30 PM- 04:00 PM  | Room 607 | 284, 602, 610, 699, 774      |
| PS8                                                                | PE-ES      | 25th Sep FRI<br>2:30 PM- 04:00 PM  | Room 608 | 249, 259, 294, 505, 609, 732 |
| PS9                                                                | CV-PR-NLP  | 25th Sep FRI<br>4:30 PM- 06:00 PM  | Room 605 | 272, 467, 546, 570, 779, 780 |
| PS10                                                               | CS-SDE     | 25th Sep FRI<br>4:30 PM - 06:00 PM | Room 606 | 143, 580, 670, 714, 736, 753 |
| PS11                                                               | RE-SPT     | 25th Sep FRI<br>4:30 PM - 06:00 PM | Room 607 | 182, 468, 575, 608, 647, 786 |
| PS12                                                               | RIA        | 25th Sep FRI<br>4:30 PM - 06:00 PM | Room 608 | 213, 307, 315, 816           |

| DAY 2: 26 SEPTEMBER 2026 • SATURDAY: PHYSICAL (ON-SITE) MORNING |            |                                   |          |                              |
|-----------------------------------------------------------------|------------|-----------------------------------|----------|------------------------------|
| ID                                                              | SHORT NAME | TIME                              | ROOM     | PAPER IDs                    |
| PS13                                                            | AIML-PS6   | 26th Sep SAT 9:00<br>AM -10:30 AM | Room 605 | 521, 542, 556, 565, 583      |
| PS14                                                            | CSA        | 26th Sep SAT 9:00<br>AM -10:30 AM | Room 606 | 186, 247, 324, 642, 721      |
| PS15                                                            | PE-MD      | 26th Sep SAT 9:00<br>AM -10:30 AM | Room 607 | 267, 330, 374, 760, 807, 812 |
| PS16                                                            | AIML       | 26th Sep SAT 9:00<br>AM -10:30 AM | Room 608 | 157, 169, 592, 516, 709      |

| DAY 1: 25 SEPTEMBER 2026 • FRIDAY- ONLINE (VIRTUAL) |            |                                    |                                                                                                                                                                 |                                   |
|-----------------------------------------------------|------------|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| ID                                                  | SHORT NAME | TIME                               | ROOM                                                                                                                                                            | PAPER IDs                         |
| VS1                                                 | AIML-VS1   | 25th Sep FRI<br>2:30 PM- 04:00 PM  | <a href="https://bdren.zoom.us/j/98632742712?pwd=77VoBcNdgqXMSCYm63rVhbVUFT5bW0.1">https://bdren.zoom.us/j/98632742712?pwd=77VoBcNdgqXMSCYm63rVhbVUFT5bW0.1</a> | 135, 153, 159, 160, 171, 274      |
| VS2                                                 | AIML-VS1   | 25th Sep FRI<br>2:30 PM- 04:00 PM  | <a href="https://bdren.zoom.us/j/93170845976?pwd=UJFCfR47zk8gDpNzeZDs3QdJxUPRW9.1">https://bdren.zoom.us/j/93170845976?pwd=UJFCfR47zk8gDpNzeZDs3QdJxUPRW9.1</a> | 355, 376, 472, 481, 506, 508      |
| VS3                                                 | AIML-VS1   | 25th Sep FRI<br>2:30 PM- 04:00 PM  | <a href="https://bdren.zoom.us/j/99144054265">https://bdren.zoom.us/j/99144054265</a>                                                                           | 286, 510, 619, 669, 781, 793      |
| VS4                                                 | AIML-IA    | 25th Sep FRI<br>4:30 PM - 06:00 PM | <a href="https://bdren.zoom.us/j/94651362051?pwd=J5dWYAnMTvdKyMs5qftb4KI5TWtw80.1">https://bdren.zoom.us/j/94651362051?pwd=J5dWYAnMTvdKyMs5qftb4KI5TWtw80.1</a> | 399, 410, 448, 679, 697, 842, 846 |

| ID  | SHORT NAME | TIME                               | ROOM                                                                                                                                                            | PAPER IDs                    |
|-----|------------|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| VS5 | PE-RE      | 25th Sep FRI<br>4:30 PM - 06:00 PM | <a href="https://bdren.zoom.us/j/99708416325?pwd=4VlhNvts9ctpUQCJi4Efp0mEX1CFF3.1">https://bdren.zoom.us/j/99708416325?pwd=4VlhNvts9ctpUQCJi4Efp0mEX1CFF3.1</a> | 241, 361, 385, 401, 434, 754 |
| VS6 | CS-DE-QC   | 25th Sep FRI<br>4:30 PM - 06:00 PM | <a href="https://bdren.zoom.us/j/95853395239?pwd=SeRLfXhyerDfVkedn9x5nRkf6jmuV0.1">https://bdren.zoom.us/j/95853395239?pwd=SeRLfXhyerDfVkedn9x5nRkf6jmuV0.1</a> | 235, 260, 302, 381, 584, 695 |

**DAY 2: 26 SEPTEMBER 2026 • SATURDAY: ONLINE (VIRTUAL)**

| ID  | SHORT NAME | TIME                              | ROOM                                                                                                                                                            | PAPER IDs                         |
|-----|------------|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| VS7 | ST-ICA     | 26th Sep SAT<br>9:00 AM -10:30 AM | <a href="https://bdren.zoom.us/j/98721132299?pwd=1LW8bUoxvNRtcJZ6127bzfBZ3xRgp6.1">https://bdren.zoom.us/j/98721132299?pwd=1LW8bUoxvNRtcJZ6127bzfBZ3xRgp6.1</a> | 196, 271, 301, 537, 577, 722      |
| VS8 | IoT-IES    | 26th Sep SAT<br>9:00 AM -10:30 AM | <a href="https://bdren.zoom.us/j/91519571555?pwd=jjSQ81pa8qNCjXBnaxbZIUeDIGq3vq.1">https://bdren.zoom.us/j/91519571555?pwd=jjSQ81pa8qNCjXBnaxbZIUeDIGq3vq.1</a> | 640, 281, 283, 311, 371, 379      |
| VS9 | ES-RCA     | 26th Sep SAT<br>9:00 AM -10:30 AM | <a href="https://bdren.zoom.us/j/95853395239?pwd=SeRLfXhyerDfVkedn9x5nRkf6jmuV0.1">https://bdren.zoom.us/j/95853395239?pwd=SeRLfXhyerDfVkedn9x5nRkf6jmuV0.1</a> | 163, 245, 251, 524, 621, 737, 826 |

**PS1 | 25th Sep FRI 09:00 AM -10:30 AM**

**TRACK: AIML-PS1 (Artificial Intelligence and Machine Learning)**

| Room / Location   |          | Room 605                                                                                                           | Session Chair(s) | Dr. Iqbal Ahmed<br>Dr. Engr. Muhibul Haque Bhuyan                                               |
|-------------------|----------|--------------------------------------------------------------------------------------------------------------------|------------------|-------------------------------------------------------------------------------------------------|
| Session Moderator |          | Kazi Md. Abrar Yeaser                                                                                              | Volunteer(s)     | Nusrath Saqi<br>Mohammad Rakibur Rahaman                                                        |
| #                 | Paper ID | Paper Title                                                                                                        |                  | Author(s)                                                                                       |
| 1                 | 32       | Vocal Tract-Inspired Speech Activity Detection: An Improved Combo-SAD Approach                                     |                  | Tanwi Chakraborty*<br>Nursadul Mamun<br>Taieba Taher                                            |
| 2                 | 261      | Cross-Circuit Generalization of Graph Neural Networks (GNN) for VLSI Fault Detection                               |                  | Rejwan Inteser<br>Istiyak Ahammed*<br>Nazmul Haque<br>Md. Hasan Maruf<br>ASM Shihavuddin        |
| 3                 | 269      | Venue aware Elo Poisson Machine Learning and Monte Carlo Forecasting of the 2026 FIFA World Cup                    |                  | Pamel Das*<br>Md. Mahmodul Haque Alve<br>Dr. Md Shamsul Arifin<br>Abir Dhar<br>Aditta Chowdhury |
| 4                 | 362      | Explainable Boosting Machine vs. Post-hoc Explainable AI: Interpretability and Performance in Academic Forecasting |                  | Tasnim Tamanna Orpa*<br>Md. Monjarul Alam<br>Md Abdullah Al Zubaer                              |
| 5                 | 544      | Variance-Guided Multi-Slice Tensor Ring Decomposition for Classification of Alzheimer's Disease from rs-fMRI       |                  | Nusrat Jahan Iqra*<br>Somen Dutta<br>Mehdi Hasan Chowdhury                                      |

| PS2   25th Sep FRI 09:00 AM -10:30 AM                          |                   |                                                                                                                              |                                                                                                                           |
|----------------------------------------------------------------|-------------------|------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| TRACK : AIML-PS2(Artificial Intelligence and Machine Learning) |                   |                                                                                                                              |                                                                                                                           |
| Room / Location                                                | Room 606          | Session Chair(s)                                                                                                             | Dr. Md. Azad Hossain<br>Dr. MD RAUF UL KARIM KHAN                                                                         |
| Session Moderator                                              | Nadim Bin Hossain | Volunteer(s)                                                                                                                 | Sayeed Ibne Saif<br>Amit Chowdhury                                                                                        |
| #                                                              | Paper ID          | Paper Title                                                                                                                  | Author(s)                                                                                                                 |
| 1                                                              | 377               | GCAF-Net: A Lightweight GSession Moderator Coordinate Attention Fusion Network for Rice Leaf Disease Classification          | Madhubi Ashraf Omi*<br>Md. Nur Alam<br>Shahmiraj Ehesan<br>Darus Salam Tasin<br>Yousuf Asgor Mumin<br>Khaladul Belal      |
| 2                                                              | 383               | EfficientMSwin: A YOLO-Guided Multi-Scale Swin Transformer for Fine-Grained Malaria Cell Stage Classification                | Obaidul Haque*<br>Mariha Tabassum<br>Ferdaus Anam Jibon<br>Tania Akter Setu<br>Shah Mostofa Tariquzzaman                  |
| 3                                                              | 421               | Seismic Risk Assessment in Chattogram: An Explainable Machine Learning Approach                                              | Sabrina Sultana<br>Samia Shafique<br>syeda akter*                                                                         |
| 4                                                              | 429               | Explainable Multi-Task Learning for Joint Prediction and Optimization of Vehicular Emissions Using Pareto Trade-off Analysis | Raiyan Alam*<br>Nusrat Jahan Mim<br>Wrishika Saha<br>Md. Redwanul Haque                                                   |
| 5                                                              | 590               | A Hybrid Deep Learning and Metaheuristic Approach for Walnut Leaf Disease Identification using SQZRFBA                       | Arin Pathan<br>Md Foysal Ahmed<br>Basarat Binte Basar<br>Mst.Shohana Akther<br>Shoriful Islam Apu<br>Sadman Samir Rafith* |

| PS3   25th Sep FRI 09:00 AM -10:30 AM                |                |                                                                                                                                               |                                                                                                          |
|------------------------------------------------------|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| TRACK : Artificial Intelligence and Machine Learning |                |                                                                                                                                               |                                                                                                          |
| Room / Location                                      | Room 607       | Session Chair(s)                                                                                                                              | Dr. Dipankar Das<br>Dr. Mohammed Moshuiul Hoque                                                          |
| Session Moderator                                    | Mahmudul Hasan | Volunteer(s)                                                                                                                                  | Mehedi Hasan<br>Tasnia Chowdury                                                                          |
| #                                                    | Paper ID       | Paper Title                                                                                                                                   | Author(s)                                                                                                |
| 1                                                    | 598            | Analyzing Modern Activation Functions to Optimize Gated Recurrent Networks for Sentiment Classification                                       | Tasnimul Afif*<br>Tahsan Uddin<br>Kazi Fazlee Rabbi                                                      |
| 2                                                    | 611            | SeverityFusionNet: Bidirectional Cross-Modal Attention for Multimodal Bangla Disaster Classification and Severity Estimation                  | Asif Khan<br>Aditi Tasmin<br>MD. ABDUL KAIUM SHOHUG*<br>Md. Noyon Islam<br>Md. Sha Jihad Uddin Chowdhury |
| 3                                                    | 627            | Beyond the Random Split: Quantifying Temporal Concept Drift in Machine Learning-Based IoT Intrusion Detection                                 | Musfiqur Rahman<br>Hasibul Hasan Rifat<br>Farhan Sadik Arnob<br>Md. Samiul Islam*                        |
| 4                                                    | 668            | Symptom-Driven Machine Learning for Early PCOS Screening in Bangladeshi Women: A Comprehensive Comparative Study with Explainability Analysis | MD. ABDUL KAIUM SHOHUG*<br>Asif Khan<br>Md. Sha Jihad Uddin Chowdhury<br>Md. Noyon Islam                 |

|   |     |                                                                                          |                                                                                                                                                                                                         |
|---|-----|------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5 | 674 | AI-Powered Personalized Product Recommendation and Decision-Making Systems in E-Commerce | MD Shahidul Islam Fakir<br>Mohammad Saddam Hosen<br>Emu Barua<br>Mohammad Tarek Aziz*<br>Sabiroy Sardor<br>Temur Eshchanov<br>Safargul Jurayeva<br>Seitnazarov Kuanishbay Kenesbaevich<br>Tanjim Mahmud |
|---|-----|------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

**PS4 | 25th Sep FRI 09:00 AM -10:30 AM**

**TRACK : Communication and Electronic Systems**

| Room / Location   | Room 608            | Session Chair(s)                                                                                                                         | Dr. Md. Selim Hossain<br>Dr. Muhammad Asad Rahman                                                                      |
|-------------------|---------------------|------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| Session Moderator | Estiak Ahamed Sazid | Volunteer(s)                                                                                                                             | Anup Kumar Dipta<br>Fahmida Sultana Rifat                                                                              |
| #                 | Paper ID            | Paper Title                                                                                                                              | Author(s)                                                                                                              |
| 1                 | 273                 | Secrecy Capacity of RIS-Assisted Radio-Frequency Networks over Generalized-Gamma Fading with Multiple Eavesdroppers                      | Md. Tanvirul Islam*<br>Md. Zahurul Islam Sarkar<br>Navid Haider                                                        |
| 2                 | 373                 | ARNet: Lightweight Attention-Guided Residual Network for Automatic Modulation Classification Across Multi-Class and Variable SNR Regimes | Nur Alam*<br>Madhubi Ashraf Omi<br>Md. Taisirul Muktadi<br>Darus Salam Tasin<br>Shahmiraj Ehesan<br>Yousuf Asgor Mumin |
| 3                 | 594                 | A Hybrid RF/THz Communication Framework with Reconfigurable Intelligent Surface: Performance Analysis                                    | Md. Faisal Reza<br>Md. Jakerul Islam<br>Md. Abdur Rakib*<br>Md. Ibrahim                                                |
| 4                 | 633                 | Subwavelength Metamaterial Cladding for Broadband and Flat Supercontinuum Generation in AlN Waveguides                                   | Zahida Zakir*<br>Jobaida Akhtar<br>Mohammad Istiaque Reja                                                              |
| 5                 | 776                 | A Q-Learning Approach to Joint Optimization of Spectrum, Interference, and Energy in CR-VANETS                                           | Md. Asif Hossain*<br>Md. Maruf Hasan<br>Israt Jahan Mumu                                                               |
| 6                 | 911                 | Suppression of Copper Diffusion in IGZO Thin-Film Using a Molybdenum Barrier Layer and HVPDA Treatment                                   | Mohammad Rabib Uddin<br>Upama Dev<br>Jarin Tasnim Heia<br>Md. Saniat Rahman Zishan<br>MD RAUF UL KARIM KHAN*           |

**PS5 | 25th Sep FRI 2:30 PM- 04:00 PM**

**TRACK : Artificial Intelligence and Machine Learning**

| Room / Location   | Room 605        | Session Chair(s)                                                                                               | Lt Col Dr. Hussain Nyeem<br>Dr. Md Abdul Wahed                   |
|-------------------|-----------------|----------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| Session Moderator | Noortaz Rezoana | Volunteer(s)                                                                                                   | Md Saiful Alam, Mohammad<br>Rakibur Rahaman                      |
| #                 | Paper ID        | Paper Title                                                                                                    | Author(s)                                                        |
| 1                 | 263             | Customer Satisfaction Analysis in Water Supply Utility Using Machine Learning: A Case Study of Chattogram WASA | Shafiqul Bashar*<br>Lutfe Jahan                                  |
| 2                 | 704             | TriFuseNet: A Three-Stream Cross-Attention Network for Few-Shot Multimodal Idiom Ranking                       | Farhana Sultana Ananna*<br>Puja Chakraborty<br>Shima Chakraborty |

|   |     |                                                                                                                                              |                                                                                      |
|---|-----|----------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| 3 | 707 | An AI-Driven Lightweight System for Smartphone-Based Beef and Buffalo Meat Authentication in Real-World Markets                              | Pratik Dav<br>Barna Dasgupta<br>Mohammad Minhaj Uddin Rafi<br>Shahid Md. Asif Iqbal* |
| 4 | 710 | Macro Regime-Conditioned Offline Reinforcement Learning with SHAP Interpretability for Emerging-Market FX Risk Management                    | Rohit Roy*<br>MD. SALAUDDIN SHAON<br>Rafid Hossain<br>Mejbah Ahammad                 |
| 5 | 719 | A Lightweight Univariate TCN-SE Framework for Battery State of Health Prediction                                                             | Bijoy Das*<br>Proshanto Das<br>Anurag Debnath                                        |
| 6 | 748 | A Performance Benchmarking of Lightweight CNNs and Vision Transformers for Automated Diabetic Retinopathy Grading Under Resource Constraints | Md. Hasibur Rahman<br>Sudipta Das<br>Samina Alam<br>Rajendra Raj Shearam<br>Joy Dey  |

**PS6 | 25th Sep FRI 2:30 PM- 04:00 PM**

**TRACK : Artificial Intelligence and Machine Learning**

| Room / Location   | Room 606          | Session Chair(s)                                                                                                                                       | Dr. Muhammad Kamal Hossen,<br>Prof. Dr. Md Fazlul Kader                                                           |
|-------------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| Session Moderator | MD. Tamim Hossain | Volunteer(s)                                                                                                                                           | Sayed Ibne Saif<br>Shovon Das Dipta                                                                               |
| #                 | Paper ID          | Paper Title                                                                                                                                            | Author(s)                                                                                                         |
| 1                 | 777               | HybridDQN: A Deep Reinforcement Learning Framework with Attention-Augmented Experience Replay for Autonomous Resource Allocation in Cloud Environments | Avijit Chowdhury*<br>SHEIKH KAMRUL HASAN OMUR<br>Khalilur Rahman<br>Md. Delwer Hossain<br>Md. Saiful Islam        |
| 2                 | 784               | Global Peptide Feature Enrichment for Protein Binding Peptide Prediction Using Machine Learning                                                        | S M Abdullah Ashik*<br>Dr. Julia Rahman<br>Md Zahim Hassan                                                        |
| 3                 | 805               | A Deployment-Oriented Data-Scarcity-Aware Multi-Dataset Evaluation of Lightweight Deep Learning Models for Waste Classification                        | Nazat E Rose Rhythm<br>Akib Hasan Pyil<br>Most. Sumaiya Habiba Nishat<br>Jafir Islam Siam<br>Kamruddin Nur*       |
| 4                 | 847               | SA-XSpam: A Sentiment-Aware Explainable Hybrid Framework for YouTube Comment Spam Detection with Semantic Robustness                                   | Abdul Hasib*<br>Md. Faishal Ahmed<br>Faiaz Ahmed<br>Arjina Nasrin Khan<br>Nasif Shahriar Alif<br>Md. Shakil Ahmed |
| 5                 | 896               | Robust Emotion Detection in Bangla, Banglish, and Bangla-Banglish Mixed Text Using Transformer Ensembles and Cross-Domain Analysis                     | Bibi Hazaratun Nesa<br>Sajjad Hosen Emon<br>Ashi Datta<br>Md. Hasan*                                              |

**PS7 | 25th Sep FRI 2:30 PM- 04:00 PM**

**TRACK : BME-HT (Artificial Intelligence and Machine Learning)**

| Room / Location   | Room 607        | Session Chair(s) | Dr. Quazi Delwar Hossain<br>Dr. Sumit Majumder |
|-------------------|-----------------|------------------|------------------------------------------------|
| Session Moderator | Mr. Somen Dutta | Volunteer(s)     | Mehedi Hasan<br>Md. Abdullah                   |
| #                 | Paper ID        | Paper Title      | Author(s)                                      |

|   |     |                                                                                                                                  |                                                                                                                              |
|---|-----|----------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| 1 | 284 | Quantitative Explainability of a Multichannel ECG-PPG CNN for Tachycardia Detection                                              | Pamel Das*<br>Md. Mahmodul Haque Alve<br>Dr. Md. Shamsul Arifin<br>Abir Dhar<br>Aditta Chowdhury                             |
| 2 | 602 | Multi-View Feature Selection Network for Automated Kidney Stone Detection in Medical Imaging                                     | Abdullah Al Alif*<br>Sabiha Jannat<br>Ferdaus Anam Jibon<br>Mohshina Sultana<br>Shah Mostofa Tariquzzaman<br>Md. Abdur Rahim |
| 3 | 610 | Mental Attention Classification Framework Using Electroencephalogram                                                             | Md. Zahid Hasan*<br>Md. Fazlul Karim Khondakar<br>Mehdi Hasan Chowdhury                                                      |
| 4 | 699 | Depression, Anxiety, Stress, and Academic Engagement: A Comprehensive Dataset of University Students                             | Md Ali Akbar Sami<br>Sakib Chowdhury<br>Puspita Bhattacharjee<br>Mohammad Hasan*<br>Mohammad Asaduzzaman Rasel               |
| 5 | 774 | A Data-Driven SEIRVD Model for Measles Transmission Dynamics in Yemen: Stability Analysis and Efficacy of Combined Interventions | Md. Arafat Chad*<br>Lincoln Podder<br>MD. Ehsanul Kabir<br>Ayaz Ahmad Khan<br>Mayesha Sharmim Tisha                          |

**PS8 | 25th Sep FRI 2:30 PM- 04:00 PM**

**TRACK : Power Engineering and Energy Systems**

| Room / Location   |          | Room 608                                                                                                                                          | Session Chair(s) | Dr. Nur mohammad<br>Dr. Mohammad Nasir Uddin                                                                               |
|-------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------|------------------|----------------------------------------------------------------------------------------------------------------------------|
| Session Moderator |          | Abir Dhar                                                                                                                                         | Volunteer(s)     | Anup Kumar Dipta<br>Mubarrat E Ishmum                                                                                      |
| #                 | Paper ID | Paper Title                                                                                                                                       |                  | Author(s)                                                                                                                  |
| 1                 | 249      | Design and Performance Analysis of a PI-Controlled DC-DC Boost Converter for Precise Voltage Regulation                                           |                  | Khairun Naher<br>Abu Talha Haque Miah*<br>Sumon Hossain<br>Miss Al Ive<br>Nawshad Ahmed Nihal                              |
| 2                 | 259      | Modeling and Simulation of a Bidirectional Single-Phase EV Charger with Interleaved Boost Converter and Phase-Shifted Full-Bridge DC-DC Converter |                  | Sadikur Rahman<br>Ariful Islam Riad*<br>Md. Hafiz Ahamed<br>Md. Sakib Hassan Chowdhury<br>Arijit Das<br>Md. Amir Hamza     |
| 3                 | 294      | Design and Comprehensive Analysis of Single Active Bridge DC-DC Converter for Solid State Transformer System                                      |                  | Sadikur Rahman<br>Ariful Islam Riad*<br>Arijit Das<br>Sarafat Hussain Abhi<br>Md. Sakib Hassan Chowdhury<br>Md. Amir Hamza |
| 4                 | 505      | Degradation-Aware Rule-Based Industrial EMS for PV-Battery Microgrids with Stateflow                                                              |                  | Utchash Dev Nath*<br>Swarup Paul<br>Jubayer Hasan Abdullah                                                                 |
| 5                 | 609      | A Smart ROCOF-Based Fast Under-Frequency Load Shedding Scheme with Local SCADA Monitoring for Low Inertia Grids                                   |                  | Anup Kamal Dipta*<br>Afra Obaidul<br>Abir Dhar<br>Kakon Nandi                                                              |

|   |     |                                                                                          |                                                                                                    |
|---|-----|------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| 6 | 732 | Supercapacitor-Based Energy Storage for Residential PV Applications: A Comparative Study | Md. Aminul Hoque*<br>Biplob Sarker<br>MD. Nahid Billah<br>Al Amin Hossain<br>Shah Mohazzem Hossain |
|---|-----|------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|

**PS9 | 25th Sep FRI 4:30 PM- 06:00 PM**

**TRACK : Computer Vision, Pattern Recognition, Natural Language Processing**

| Room / Location   |          | Room 605                                                                                                                                         | Session Chair(s) | Dr. Md. Selim Hossain<br>Dr. Tanjim Mahmud                                                                           |
|-------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------|------------------|----------------------------------------------------------------------------------------------------------------------|
| Session Moderator |          | Md. Hasan                                                                                                                                        | Volunteer(s)     | Nusrath Saqi<br>Mohammad Rakibur Rahaman                                                                             |
| #                 | Paper ID | Paper Title                                                                                                                                      |                  | Author(s)                                                                                                            |
| 1                 | 272      | ClusterFlow: A Temporal Measurement-Based Quantum Computing Simulation Framework for Resource-Optimized and Loss-Aware Cluster-State Subroutines |                  | Arnob Aich*<br>Shamiul Islam<br>Md. Nasif Rafidi<br>Md. Sadman Hossain,<br>Nuren Tasnim Risha<br>Samiha Islam Abrita |
| 2                 | 467      | An Empirical Study of Input Resolution and Sliced Inference for Aerial Human and Vehicle Detection                                               |                  | Md Dip*<br>Md. Shakibul Islam Shakib<br>Ferdaus Anam Jibon<br>Sourav Simanta<br>Mohshina Sultana                     |
| 3                 | 546      | A Multi-Level Deep Learning Framework to Classify Mental Health Disorders with XAI                                                               |                  | Payel Sen*<br>Muhammad Ibrahim Khan                                                                                  |
| 4                 | 570      | HybridIndic-Net: A Compact Convolutional-Stem Vision Transformer for Handwritten Indic Script Recognition                                        |                  | Pranta Chowdhury<br>Tasnim Akter<br>Avishek Chowdhury*<br>Md. Imtiaz Uddin                                           |
| 5                 | 779      | Chatga-Boli: A Data-Centric Framework for Bangla Dialect Translation                                                                             |                  | Mohammed Parves*<br>Manoara Begum<br>Mohammed Shadman                                                                |
| 6                 | 780      | ADR-Net: Adaptive Dual-Path Routing for Lightweight Image Denoising                                                                              |                  | Avijit Chowdhury*<br>Khalilur Rahman<br>Tasnuha Hasan<br>Md Samir Murad<br>Md. Saiful Islam                          |

**PS10 | 25th Sep FRI 4:30 PM - 06:00 PM**

**TRACK : Cybersecurity, Software and Data Engineering**

| Room / Location   |          | Room 606                                                                                                 | Session Chair(s) | Dr. Mohammad Nurul Huda<br>Dr. Nursadul Mamun              |
|-------------------|----------|----------------------------------------------------------------------------------------------------------|------------------|------------------------------------------------------------|
| Session Moderator |          | Rashed Miah                                                                                              | Volunteer(s)     | Sayed Ibne Saif<br>Amit Chowdhury                          |
| #                 | Paper ID | Paper Title                                                                                              |                  | Author(s)                                                  |
| 1                 | 143      | Critical-Boundary Sensitivity Analysis of a Lifecycle-Aware IoT Threat-Scoring Framework on BoT-IoT      |                  | Purba Mojumdar*                                            |
| 2                 | 580      | Detecting Obfuscated Malware in Volatile Memory: Classification and Plugin-Level Explainability via SHAP |                  | Tasrin Noman Mifta*<br>HRIDIKA CHANDA<br>Kazi Fazlee Rabbi |

|   |     |                                                                                                                                                         |                                                                                                                                    |
|---|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| 3 | 670 | Advanced Stock Price Forecasting: A Comparative Analysis of various Models for Tesla                                                                    | Nazma Akther*<br>Avisheak Das<br>Mohammed Saifuddin Munna                                                                          |
| 4 | 714 | A Web-Based University Thesis Management System with Capacity-Constrained Supervisor Assignment                                                         | MD Abdullah Jobayer*<br>MD Reaz Uddin Chowdhury<br>Amamul Ahasan<br>Dr. Shahid Md. Asif Iqbal<br>Mohammad Hasan                    |
| 5 | 736 | CORAL-Regularized Cross-Dataset Generalization for Network Intrusion Detection Under Native Feature-Schema Mismatch: A Rigorously Validated Null Result | Abdul Hasib*<br>Nasif Shahriar Alif<br>Arjina Nasrin Khan<br>Md. Ashiqussalehin<br>Md. Rakibul Hasan Nishat<br>D. M. Khalid Mahmud |
| 6 | 753 | A Decentralized LSTM-based Detection and Mitigation Framework for FDI Attacks in Isolated Microgrids                                                    | Madhurima Das*<br>Sumaya Tasnim<br>Emon Chowdhury<br>Siam Billah<br>Subrata K. Sarker                                              |

**PS11 | 25th Sep FRI 4:30 PM - 06:00 PM**

**TRACK : Renewable Energy and Sustainable Power Technologies**

| Room / Location   |          | Room 607                                                                                                                                         | Session Chair(s)                                                                                            | Dr. Kazi Tanvir Ahmmed, Dr. Md. Tofael Ahmed |
|-------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|----------------------------------------------|
| Session Moderator |          | Ms. Tashin Hossain                                                                                                                               | Volunteer(s)                                                                                                | Mehedi Hasan<br>Tasnia Chowdury              |
| #                 | Paper ID | Paper Title                                                                                                                                      | Author(s)                                                                                                   |                                              |
| 1                 | 182      | Techno-Economic Analysis and Design Optimization of a Solar-Powered Electric Vehicle Charging Station Along the Dhaka - Chittagong Highway       | Golam Rasul*<br>Md. Ashraful Hoque Rupom<br>Md. Manjurul Gani<br>Md. Adnan Faisal Siddique<br>Nur Mohammad  |                                              |
| 2                 | 468      | Operational Analysis of Backup and Grid-Supplied Charging Modes in a 214 kW Solar-Grid Hybrid EV Charging Infrastructure                         | Md. Towhid Hossain*<br>Sheik Mohammad Ali Hossen Foyzal<br>Khandakar Abdullah Al Mamun                      |                                              |
| 3                 | 575      | Transfer Learning with Conformalized Quantile Regression for Solar PV Forecasting Under Load-Shedding-Driven Data Scarcity                       | Rakib Abdullah<br>K.M. Tahlil Mahfuz Faruk*                                                                 |                                              |
| 4                 | 608      | A Flood-Adaptive Agrivoltaic System for Year Round Production: Integrating Floating Seedling Banks and Indigenous Crop Cultivation in Bangladesh | Md Shoab Khan*<br>Md Saqlain Mustak Saki<br>Md Hujaifa Ahmmed Tanvir<br>ASM Shihavuddin                     |                                              |
| 5                 | 647      | Explainable Hybrid Deep Learning Framework for Multi-Location Short-Term Photovoltaic Energy Forecasting                                         | Juhi Jannat Mim*<br>Md. Nazmul Md. Nazmul<br>Robiul Hossen<br>Md. Aminul Islam Pallab<br>Asiqur Rahman Abir |                                              |
| 6                 | 786      | Dynamic 24-Hour Voltage Stabilization and Loss Minimization in IEEE 14-Bus System via PSO-Optimized Capacitor and Battery-EV Coordination        | Joyantha Sarkar*<br>Abir Dhar<br>Mohammad Jobaed Hossain<br>Abu Jaber                                       |                                              |

**PS12 | 25th Sep FRI 4:30 PM - 06:00 PM**

**TRACK : Robotics, IoT and Intelligent Automation**

| Room / Location | Room 608 | Session Chair(s) | Dr. Aseef Iqbal<br>Dr. Tanzilur Rahman |
|-----------------|----------|------------------|----------------------------------------|
|-----------------|----------|------------------|----------------------------------------|

| Session Moderator |          | Chowdury Fariha Kamrul                                                                                                         | Volunteer(s) | Anup Kumar Dipta<br>Fahmida Sultana Rifat                                                                                                               |
|-------------------|----------|--------------------------------------------------------------------------------------------------------------------------------|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| #                 | Paper ID | Paper Title                                                                                                                    |              | Author(s)                                                                                                                                               |
| 1                 | 213      | FreeRTOS-Based Real-Time Parking System with Mobile Application Integration                                                    |              | Ifty Imam Bin Razzak<br>Md. Zunaid Hossain<br>Saurav Chandra Das*<br>Pritom Paul                                                                        |
| 2                 | 307      | Sim-to-Sim Transfer of Reinforcement Learning-Based Quadruped Locomotion: A Transferability Study from Isaac Gym to MuJoCo     |              | MD Ashfakul Karim Kausik*<br>SK Naymul Siddque<br>Md. Sayem Hossain Bhuiyan                                                                             |
| 4                 | 315      | Learning-Augmented Control for Quadrotor MPC in MuJoCo: When a Feedforward Residual Architecture Helps, Hurts, or Does Nothing |              | MD Ashfakul Karim Kausik*<br>SK Naymul Siddque<br>Md. Sayem Hossain Bhuiyan                                                                             |
| 5                 | 816      | A Low-Cost 4-DOF Tele-Operated Robotic Arm Achieving Millimeter-Scale Precision for SME Automation                             |              | Indra Jit Barua<br>Nazia Sultana Plabon*<br>Ashok Pal Emon<br>Mohammed Saifuddin Munna<br>Joy Dey<br>Minhaz Uddin hasan<br>Anik Datta<br>Sudipto Chakma |

**PS13 | 26th Sep SAT 9:00 AM -10:30 AM**

**TRACK : Artificial Intelligence and Machine Learning**

| Room / Location   |          | Room 605                                                                                                                                                          | Session Chair(s) | Dr. MD RAUF UL KARIM KHAN<br>Dr. Mehdi Hasan Chowdhury                                                         |
|-------------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|----------------------------------------------------------------------------------------------------------------|
| Session Moderator |          | Avisheak Das                                                                                                                                                      | Volunteer(s)     | Susmoy Barua<br>Md Abdullah                                                                                    |
| #                 | Paper ID | Paper Title                                                                                                                                                       |                  | Author(s)                                                                                                      |
| 1                 | 521      | How Much Prosody Is Enough? Compact Prosody Bottlenecks for Lightweight Text-to-Speech                                                                            |                  | Obaidul Haque*<br>Abrar Rakin<br>Ferdaus Anam Jibon<br>Esrat Nourin<br>Mohshina Sultana                        |
| 2                 | 542      | A Comparative Analysis of Swin Transformer and Hybrid CNN-Transformer Architectures for Real-Field Banana Leaf Disease Classification Using a Bangladeshi Dataset |                  | Arnab Paul John*<br>Amit Paul Joy<br>Nadim Bin Hossain                                                         |
| 3                 | 556      | A Reliability Evaluation Protocol for Graph Neural Networks under Structural Distribution Shift: Calibration, Robustness, and Explanation Fidelity                |                  | Dipta Prattoy Karmakar*<br>Mohammad Mahmudul Hasan                                                             |
| 4                 | 565      | Morphology-Informed Hybrid Deep Learning for Brugada Syndrome Detection from 12-Lead ECG                                                                          |                  | Tanvir Shahadat Shuva*<br>Nayeem Uddin<br>Md Zakir Hossen<br>Abbas Uddin Chowdhury<br>Dr. Ayesha Ahmed Khan    |
| 5                 | 583      | ViT-Dominant Gated Efficient Hybrid Network for Jute Leaf Disease Classification                                                                                  |                  | Abu Bakar Rakib<br>Obaidul Haque*<br>Ferdaus Anam Jibon<br>Dr. Md. Ashraful Islam<br>Md. Mahamudul Hasan Rahad |

**PS14 | 26th Sep SAT 9:00 AM -10:30 AM**

| TRACK : Control Systems and Automation |                                  |                                                                                                                       |                                                                                                                                               |
|----------------------------------------|----------------------------------|-----------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Room / Location                        | Room 606                         | Session Chair(s)                                                                                                      | Dr. Sajal Chandra Banik<br>Dr. Sheikh Md. Rabiul Islam                                                                                        |
| Session Moderator                      | Mohammad Rezaur Rahman Chowdhury | Volunteer(s)                                                                                                          | Vicky Nandy<br>Binoy                                                                                                                          |
| #                                      | Paper ID                         | Paper Title                                                                                                           | Author(s)                                                                                                                                     |
| 1                                      | 186                              | Quantum-Inspired Actor-Critic Reinforcement Learning for Autonomous Drone Navigation in High-Obstacle Smart Cities    | Sejuti Saha*<br>Ratul Bhattacharjee<br>Nasiba Nishat<br>Tamjid Muttaqi                                                                        |
| 2                                      | 247                              | Dynamic Modeling of Electric Power Steering: Energy Consumption and Steering Feel Investigation                       | Robin Barua*<br>Nur Mohammad<br>Riton Kumer Das<br>Mohammad Iftekher Uddin                                                                    |
| 3                                      | 324                              | Design and Hardware Implementation of an Analog PID Controller for Active Vibration Isolation in UAV Payloads         | Md. Noyon Islam*<br>Ragib Rownak<br>Prangon Das<br>Md Zunaid Hossen<br>Md. Irfanul Haque Siam<br>Shimanta Nandi<br>Md. Rafiul Islam           |
| 4                                      | 642                              | Design and Experimental Validation of a Low-Cost IoT-Based LPG Leakage Detection System with Local and Remote Alerts  | Intisar Jaman Chowdhury*<br>Tamim Hasan Apurbo<br>Niladri Roy Nion<br>T. A Nahian<br>Mahabub Hossain<br>Sahadat Hossain<br>Md. Mortuza Ahmmed |
| 5                                      | 721                              | Decentralized Multi-Agent Reinforcement Learning for Online ADRC Tuning in a Renewable-Integrated Shipboard Microgrid | Md. Rakkibul Islam Prottoy<br>Mst. Nusrat Jahan Asamony<br>Mahmud Hasan Shihab<br>Siam Billah<br>Md. Meraj Ahmed<br>Subrata K. Sarker*        |

| PS15   26th Sep SAT 9:00 AM -10:30 AM      |                |                                                                                                                     |                                                                                                                        |
|--------------------------------------------|----------------|---------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| TRACK : Power Electronics and Motor Drives |                |                                                                                                                     |                                                                                                                        |
| Room / Location                            | Room 607       | Session Chair(s)                                                                                                    | Dr. Nipu Kumar Das<br>Dr. Md. Istiaque Reja                                                                            |
| Session Moderator                          | Md Toukir Shah | Volunteer(s)                                                                                                        | Md Mostafa Wasif<br>Fahmida Sultana Rifat                                                                              |
| #                                          | Paper ID       | Paper Title                                                                                                         | Author(s)                                                                                                              |
| 1                                          | 267            | A Circular Energy Approach: Combining Solar Energy and Textile Waste Pyrolysis for Marine Transport in Bangladesh.  | Sanjoy Dutta*<br>Pramit Dev                                                                                            |
| 2                                          | 330            | An Oscillation-Free AC Motor Drive Using an Auxiliary Commutation Control Scheme                                    | Ariful Islam Riad*<br>Sadikur Rahman<br>Md Zunaid Hossen<br>Arijit Das<br>Md. Sakib Hassan Chowdhury<br>Md. Amir Hamza |
| 3                                          | 374            | Quantitative Evaluation of Seasonal PM <sub>2.5</sub> Influence on Utility-Scale Solar PV Performance in Bangladesh | Md Imran Hasan Emon*<br>Liton Islam<br>Shafiullah Lami<br>MD Alimul Hossain                                            |

|   |     |                                                                                                                             |                                                                                                    |
|---|-----|-----------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
|   |     |                                                                                                                             | Iftekhar Ahmmed<br>Ashadullah Al Galib                                                             |
| 4 | 760 | Blockchain-Assisted Peer-to-Peer Energy Trading for Residential Microgrids: A Technical and Economic Performance Evaluation | Abu Jaber*<br>Sathi Saha<br>Abir Dhar<br>Sumi Saha                                                 |
| 5 | 807 | Design and Analysis of an Optimized Standalone ON-Grid Hybrid Power System On MRT Dhaka, Bangladesh                         | Lincoln Podder*<br>Md. Arafat Chad<br>SHIFAIL HAQUE SHIFAT<br>Md Robiul Hossain<br>Akib junide     |
| 6 | 812 | Feasibility of a Shared Solar-Grid Hybrid Cooking Infrastructure for High-Density Residential Buildings                     | Anup Dipta*<br>Afra Obaidul<br>Shakawat Irfan<br>Fahmida Priya<br>Joyantha Sarkar<br>Tuton Mallick |

**PS16 | 26th Sep SAT 9:00 AM -10:30 AM**

**TRACK : AIML (Artificial Intelligence and Machine Learning)**

| Room / Location   |          | Room 608                                                                                                                                                 | Session Chair(s)                                                                                                                                                                          | Dr. Mohammad Nurul Huda<br>Dr. Md. Tofael Ahmed |
|-------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|
| Session Moderator |          | Not specified                                                                                                                                            | Volunteer(s)                                                                                                                                                                              | Sumaiya Khanam<br>Jubaida Khanam Juli           |
| #                 | Paper ID | Paper Title                                                                                                                                              | Author(s)                                                                                                                                                                                 |                                                 |
| 1                 | 157      | Controlled Evaluation of Input Representation and Haplotype Sorting for CNN-Based Selective Sweep Detection                                              | Md Anjir Jaman<br>Koushik Biswas Arko<br>Arizit Chaki Artha*<br>Nusrat Jahan<br>S. F. Sheikh Saadi<br>Shahriar Sadib<br>Md. Fahim Shahriar<br>Md. Mortuza Ahmmed                          |                                                 |
| 2                 | 169      | Low-Cost Vision-Based Gesture-to-Grasp Control of a 3D-Printed Robotic Hand Using MediaPipe Landmark Tracking                                            | Sadia Mannan<br>Maruf Ahmad<br>Riaz Ahammed<br>Mohiuddin Ahmad*                                                                                                                           |                                                 |
| 3                 | 592      | ChestX-Lite-KD: A Novel Deployment-Aware Distillation Framework Unifying Compression, Cross-Dataset Auditing, and Uncertainty for Tuberculosis Screening | Sadia Tasnima Shemu<br>Tanvir Alam Khondaker<br>Md. Musfiqur Rahman Akib*                                                                                                                 |                                                 |
| 4                 | 516      | Explainable Machine Learning-Based Decision Support Framework for Early Detection of Common Infant Illnesses                                             | Radoanul Arifen*<br>S M Meriyan Islam<br>Aporba Saha<br>Anisa Azad<br>Rakibul Hasan<br>Sabiqun Nahar Ritu                                                                                 |                                                 |
| 5                 | 709      | Beyond Random Validation: Drift-Aware Uncertainty Learning for Reliable Long-Term PM <sub>2.5</sub> Forecasting under Temporal Distribution Shift        | Tanvir Ahmed*<br>Sharmin Jahan Shithy<br>Md Asraful Islam Jony<br>Azizur Rahman<br>Tasnimul Ibtisham Tamim<br>Md Mukul Hossain<br>Md Mredul Hasan<br>Shahriar Sadib<br>Md. Mortuza Ahmmed |                                                 |

**VS1 | 25th Sep FRI 2:30 PM- 04:00 PM**

**TRACK : AIML-VS1(Artificial Intelligence and Machine Learning)**

|                          |                             |                         |                                                      |
|--------------------------|-----------------------------|-------------------------|------------------------------------------------------|
| <b>Room / Location</b>   | <b>Room 610 (Virtual)</b>   | <b>Session Chair(s)</b> | <b>Dr. Mohiuddin Ahmad<br/>Dr. Md. Selim Hossain</b> |
| <b>Session Moderator</b> | <b>Kafayet Monoar Nahin</b> | <b>Volunteer(s)</b>     | <b>Susmoy Barua<br/>Syed Sorfuddin Alif</b>          |

**Joining Information (Zoom):**

<https://bdren.zoom.us/j/98632742712?pwd=77VoBcNdggXMSCYm63rVhbVUfT5bWo.1>

| # | Paper ID | Paper Title                                                                                                                             | Author(s)                                                                                                                |
|---|----------|-----------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| 1 | 135      | Machine Learning-Based Aftershock Prediction Using Physics-Informed Stacked Ensembles and Magnitude-Adaptive Thresholds                 | Khondaker Iffti Hasan Turjo<br>Jarin Tasnim<br>Md. Junaid Hossain Noman<br>Arjun Kumar Bose Arnob*<br>Kamruddin Nur      |
| 2 | 153      | Neuron Activation Frequency Divergence: A Label-Free Transferability Estimator for Pretrained Deep Neural Networks                      | Rashid Tanzil*<br>Fattah Bin Siddique<br>Rafidul Islam<br>Abday Mohaimin                                                 |
| 3 | 159      | A Semi-Supervised Deep Learning Approach for Intracranial Hemorrhage Detection and Segmentation Using U-Net Variants and Hybrid ResUNet | Sadia Islam<br>MD KAVIUL HOSSAIN*                                                                                        |
| 4 | 160      | Adaptive Multi-View Ensemble for Early Preeclampsia Detection in Low-Resource Clinical Environment                                      | Aniqua Nawar<br>MD KAVIUL HOSSAIN*                                                                                       |
| 5 | 171      | Explainable Lightweight Deep Learning Framework for Large-Scale Multi-Surface Garbage Classification                                    | Md Al Jubair*<br>Tahmid Hossain<br>Fatema Binte Hassan<br>Mohammad Mehadi Hasan<br>Sumon Biswas<br>Mahfuzulhoq Chowdhury |
| 6 | 274      | Deep Learning for Precision Agriculture: Benchmarking Vision Transformers and ConvNeXt for Multi-Class Vegetable Classification         | Shahadat Hossain*<br>Shraboni Saha Puja<br>Mahamudul Hasan Medul                                                         |

**VS2 | 25th Sep FRI 2:30 PM- 04:00 PM**

**TRACK : AIML-VS1(Artificial Intelligence and Machine Learning)**

|                          |                                 |                         |                                                             |
|--------------------------|---------------------------------|-------------------------|-------------------------------------------------------------|
| <b>Room / Location</b>   | <b>Room 610 (Virtual)</b>       | <b>Session Chair(s)</b> | <b>Dr. KAMRUDDIN MD. NUR<br/>Dr. Mahfuza Farooque Mimmi</b> |
| <b>Session Moderator</b> | <b>Tahiat Mahabub Chowdhury</b> | <b>Volunteer(s)</b>     | <b>Hameem Hossain<br/>Ifty Anwar</b>                        |

**Joining Information (Zoom):**

<https://bdren.zoom.us/j/93170845976?pwd=UJFCfR47zk8gDpNzeZDs3QdJxUPRW9.1>

| # | Paper ID | Paper Title                                                                                                                                       | Author(s)                                                                         |
|---|----------|---------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| 1 | 355      | Integrated Machine Learning for Hydrate Formation Temperature Prediction, Uncertainty Quantification, and Flow Assurance in Natural Gas Pipelines | Soraiya Sayeed Mona*<br>Md. Mizanur Rahman<br>Jannatul Karimoon                   |
| 2 | 376      | AI-Generated Text Detection Using Model-Aware Stylometric Drift and Explainable AI                                                                | Md. Wahid Ferdous<br>Yucca Das<br>Abdul Asad<br>Al Arman Ovi*<br>Md. Mejbah Uddin |

| # | Paper ID | Paper Title                                                                                                                                                              | Author(s)                                                                                                                                                                          |
|---|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   |          |                                                                                                                                                                          | Nahidul Islam<br>Abrar Siddique                                                                                                                                                    |
| 3 | 472      | Explainable Hybrid CNN--Transformer Model with Balanced Cross-Attention Fusion for Autism Spectrum Disorder Detection from Facial Images                                 | Hasan Mahadi*<br>Arif Foysal Bin Haider<br>Sadnan Jamiul Haque<br>Md. Aseer Ishraq Tasin                                                                                           |
| 4 | 481      | A Gated Three-Stream Fusion Framework for Automated Brain-Stroke Classification from MRI                                                                                 | Mugdha Farheen Kibria<br>Zarin Tasnim Vega<br>Ahmad Farruk Towsif<br>MD. SAMIUL ISLAM Samiul<br>Azizul kabir jayed Jayed<br>N M Saif Kabir<br>Abdus Sium<br>Tauhidur Rahman Sakib* |
| 5 | 506      | Machine Learning-Based Classification of Aquaculture Water Quality Using Physicochemical Parameters with Explainable AI Analysis: A SHAP and LIME Interpretability Study | Anay Kumer Ghosh*<br>Ferdaus Anam Jibon<br>Wayshikur Rahman<br>Esrat Nourin<br>Md. Ashraful Islam                                                                                  |
| 6 | 508      | Trust-Gated Capability Control: Breaking the Trust–Vulnerability Paradox in Multi-Agent LLM Systems                                                                      | MD.Mehedi Nipu*<br>Chinmoy Mitra<br>Israt Moyeen Noumi<br>Tarannum Ahmed Nowshin                                                                                                   |

**VS3 | 25th Sep FRI 2:30 PM- 04:00 PM**

**TRACK : AIML-VS1(Artificial Intelligence and Machine Learning)**

|                          |                           |                         |                                                       |
|--------------------------|---------------------------|-------------------------|-------------------------------------------------------|
| <b>Room / Location</b>   | <b>Room 610 (Virtual)</b> | <b>Session Chair(s)</b> | <b>Dr. Kazi Tanvir Ahmmed<br/>Dr. Md Aminul Islam</b> |
| <b>Session Moderator</b> | <b>Md. Raisul Islam</b>   | <b>Volunteer(s)</b>     | <b>Sumaiya Khanam<br/>Rehnuma Ahmed</b>               |

**Joining Information (Zoom):** <https://bdren.zoom.us/j/99144054265>

| # | Paper ID | Paper Title                                                                                                                                 | Author(s)                                                                                                                                                                     |
|---|----------|---------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | 286      | A Two-Stage Deep Learning Architecture for MRI-Based Brain Tumor Analysis                                                                   | Piklu Datta*<br>Mrithunjoy Das Shiblu<br>Tanjim Mahmud<br>Md. Faisal Bin Abdul Aziz<br>Bekmirzayev Eshkuvvat<br>Anorgul Ashirova<br>Jushkinbek Yuldoshev<br>Yuldasheva Gulora |
| 2 | 510      | Explainable Multi-Class Mental Health Classification: A Comparative Study of Traditional Machine Learning and BERT with SHAP Explainability | Yucca Das*<br>Abdul Asad<br>Sabyasachi Chakraborty<br>Nahidul Islam<br>Abrar Siddique<br>Sakibul Islam Sunny<br>Al Arman ovi                                                  |
| 3 | 619      | FUZZLE: A Hybrid Fuzzy Ensemble Machine Learning Framework for Interpretable Alzheimer's Disease Staging Using OASIS Dataset                | Nafees Al Mahadi*<br>Fairose Chahara Tohfa<br>Syed Khairul Islam<br>Amena Afrin Alo<br>Al Mamun Samrat<br>Sizan Md Thowhidul Islam                                            |

| # | Paper ID | Paper Title                                                                                                                                | Author(s)                                                                |
|---|----------|--------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| 4 | 669      | Analyzing Global Electric Vehicle Trade Efficiency Using Machine Learning and Explainable AI                                               | Tashida Yeasmin*<br>Ismot Ara                                            |
| 5 | 781      | Exploring Metabolic Subgroups and Algorithmic Fairness in a Bangladeshi Type-2 Diabetes Cohort: An Explainable Machine Learning Case Study | Farhan Onu*<br>Jugal Krishna Das<br>Sunirmol Mollik                      |
| 6 | 793      | RAI-FedCardio: A Personalized Gated Wide-and-Deep Federated Framework with XAI and Responsible AI for Heart Disease Prediction             | Amit Hasan Sikder*<br>Lipa Akter<br>Maimona Faiza Monisha<br>Sumya Akter |

**VS4 | 25th Sep FRI 4:30 PM - 06:00 PM**

**TRACK : AIML-IA(Artificial Intelligence, Machine Learning and Intelligent Applications)**

| Room / Location   | Room 610 (Virtual) | Session Chair(s) | Dr. Md. Najmul Hossain<br>Dr. Saiful Islam |
|-------------------|--------------------|------------------|--------------------------------------------|
| Session Moderator | Yakinur Rahman     | Volunteer(s)     | Susmoy Barua<br>Syed Sorfuddin Alif        |

**Joining Information (Zoom):**

<https://bdren.zoom.us/j/94651362051?pwd=J5dWYAnMTvdKyMs5qftb4Kl5TWtw80.1>

| # | Paper ID | Paper Title                                                                                                                                     | Author(s)                                                                                                                                                                                |
|---|----------|-------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | 399      | DR-LiteNet: A Lightweight Explainable Hybrid CNN Framework for Imbalanced Diabetic Retinopathy Grading via Adaptive SMOTE Fusion                | Nishi Kanta Paul*<br>Md Shihabul Islam Shovo<br>Audrija Chowdhury<br>Israt Jahan Esha                                                                                                    |
| 2 | 410      | Evaluating Handcrafted Features and Gradient-Boosted Classifiers for AI-Generated Image Detection                                               | Syed Mehedi Hasan Nirob*<br>Moqsadur Rahman<br>Shamim Ehsan<br>Summit Haque                                                                                                              |
| 3 | 448      | Attention-Weighted Modality Fusion for Music Semantic IDs                                                                                       | Soumya Ranjan<br>Honey Dinesh Nikam*                                                                                                                                                     |
| 4 | 679      | Multi-Class Stress Level Detection Using Physiological Signals and Self-Reported Behavioral Features                                            | Nusura Nur Nowrin<br>Sohain Tabassum Biva<br>Tanjim Mahmud*<br>Md Akash Rahman<br>Mohammad Shahadat Hossain<br>Abubokor Hanip<br>Sardor Sabirov<br>Khayrulla Urozboev<br>Odilbek Kosimov |
| 5 | 697      | Generating Bangla News Headlines Using Large Language Models                                                                                    | Jakia Jannat<br>Nabidnur Abrar<br>Koushik Rudra Kanan<br>Mohammad Tahidul Kabir Tasin<br>Md Junayed<br>Muhammad Minhazul Haque<br>Bhuiyan<br>Munware Alam Nirjhor<br>Debabrata Mallick*  |
| 6 | 842      | Biochemical Interaction-Driven Ensemble Classification of Gallstone Disease with Progressive Feature Ablation and CRP-VitD Attribution Analysis | Md Samin Yeasher*<br>Rifazul Karim Rahat<br>Al Sharia                                                                                                                                    |

| # | Paper ID | Paper Title                                                                                        | Author(s)                                                                                                                   |
|---|----------|----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| 7 | 846      | Learning from Sparse Inspection Logs: An Explainable Framework for Manufacturing Defect Prediction | Md Sarfuddin Anowar Khan*<br>Mohammed Jubaed<br>Ikbal Hossain<br>Aiman Ibn Hashem<br>Syed Tahmid Jamil<br>Md Ariful Ferdous |

#### VS5 | 25th Sep FRI 4:30 PM - 06:00 PM

TRACK : PE-RE(Power Engineering , Renewable Energy and Sustainable Power Technologies)

|                   |                    |                  |                                                |
|-------------------|--------------------|------------------|------------------------------------------------|
| Room / Location   | Room 610 (Virtual) | Session Chair(s) | Dr. Md. Mahbub Hasan<br>Prof. Dr. Nur Mohammad |
| Session Moderator | Mehrin Islam       | Volunteer(s)     | Hameem Hossain<br>Ifty Anwar                   |

#### Joining Information (Zoom):

<https://bdren.zoom.us/j/99708416325?pwd=4VihNvts9ctpUQCJi4EfpomEX1CFF3.1>

| # | Paper ID | Paper Title                                                                                                                                                                                                    | Author(s)                                                                                                       |
|---|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| 1 | 241      | Computational Analysis, Performance Enhancement and Comparison of Cs <sub>2</sub> SnBr <sub>6</sub> , ZnSnN <sub>2</sub> , Cs <sub>2</sub> SnBr <sub>6</sub> /ZnSnN <sub>2</sub> -based Perovskite Solar Cells | Nadira Farjana*<br>Mst. Jahida Akter Jim<br>Rabeya Khan<br>Md. Shizer Rahman                                    |
| 2 | 361      | Assessment of Solar PV-Battery Based micro-grid for Sustainable Power Supply in Bhashan Char                                                                                                                   | Sagor Biswas*<br>Shah Mohazzem Hossain                                                                          |
| 3 | 385      | A Curtailment Aware Adaptive Midday Discount Tariff for a High-Solar Feeder: Case Study of a 433.4 kWp On-Grid Rooftop PV System                                                                               | Partha Nath Robin*<br>Fahim Shabab<br>Sancoy Barua<br>Md. Adnan Faisal Siddique                                 |
| 4 | 401      | Enhancement of the Efficiency of Perovskite Solar Cell using MASnBr <sub>3</sub> / CsPb <sub>0.625</sub> Zn <sub>0.375</sub> I <sub>2</sub> Cl Double Layer Absorber                                           | Md. Sadik Hossain*<br>Md. Faruk Hossain                                                                         |
| 5 | 434      | Techno-Economic Analysis of a Hybrid Renewable Microgrid for Saint Martin's Island                                                                                                                             | Md Momin Hossain<br>Usha Akter<br>SHAFIQU L ISLAM*<br>Md Rasel Sarkar<br>Sanzida Akter Jeba<br>Khalid Bin Oalid |
| 6 | 754      | Renewable Hydrogen Microgrid for Ecologically Sensitive Coral Islands: A Techno-Economic and Environmental Assessment                                                                                          | Jiad Bin Asad*<br>Md. Mehedi Hasan Tusar                                                                        |

#### VS6 | 25th Sep FRI 4:30 PM - 06:00 PM

TRACK : CS-DE-QC(Cybersecurity, Data Engineering & Quantum Computing)

|                   |                    |                  |                                                    |
|-------------------|--------------------|------------------|----------------------------------------------------|
| Room / Location   | Room 610 (Virtual) | Session Chair(s) | Dr. MD RAUF UL KARIM KHAN<br>Dr. Md. Selim Hossain |
| Session Moderator | Rowshon Akter      | Volunteer(s)     | Sumaiya Khanam<br>Rehnuma Ahmed                    |

#### Joining Information (Zoom):

<https://bdren.zoom.us/j/6187750035?pwd=aafboTQkjuocMow0M2HgAOSfW9XGY6.1&omn=92094776084>

| # | Paper ID | Paper Title                                                                                                                                             | Author(s)                                                                                                                                                                                     |
|---|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | 235      | SQL-Based Association Rule Mining for Identifying Heart Disease Risk Factors: A Clinical Data Mining Approach on the Cleveland Dataset                  | Fathma Khatun*<br>Saad Ahmed                                                                                                                                                                  |
| 2 | 260      | Contract-to-Certificate: Auditable Quality Guarantees for Noisy Quantum Computation                                                                     | Shaikh Shehab Ahamed*<br>Arijit Roy<br>Arindol Paul<br>Safwan Ahmed Efaj                                                                                                                      |
| 3 | 302      | Empirical Evaluation of Zero Trust Architecture Against Lateral Movement: A Linux Namespace Testbed Study                                               | Joya Biswas*<br>Rutaban Jania<br>MD Fahim Hossain                                                                                                                                             |
| 4 | 381      | DeepGuardQR: A Hybrid Multimodal Framework for QR Phishing Detection Using DistilBERT-BiLSTM, ViT, and Handcrafted URL Features                         | Indrani Sen Toma*<br>Md. Ragib Hasan<br>Md. Asadulla Hel Galib<br>Shariful Islam<br>Amit Hasan Tonmoy                                                                                         |
| 5 | 584      | HiTrust: A Temporally Aware, Reputation Guided Federated Defense for Imbalanced Drone Intrusion Detection                                               | Md. Shafiqul Islam<br>POBITRO KUMER<br>Ilham Rahman Faysal<br>Ashniha Rashid<br>Md. Oleullah<br>Md. Alamgir Hossain*                                                                          |
| 6 | 695      | Age stratified COVID 19 Mortality, Gender Disparity and Demographic Correlates in Ahmednagar District, Maharashtra: An Integrated Epidemiological Study | Debabrata Mallick*<br>Mohammad Tahidul Kabir Tasin<br>Mohammad Al-amin Mojumder<br>Mishkatul Jannath<br>Md Junayed<br>Abdullah Al Mamun<br>Mohammed Mahmudur Rahman<br>Engr. Md Mostafa Jamal |

**VS7 | 26th Sep SAT 9:00 AM -10:30 AM**

**TRACK : ST-ICA(Smart Technologies and Intelligent Computing Applications)**

|                          |                             |                         |                                                          |
|--------------------------|-----------------------------|-------------------------|----------------------------------------------------------|
| <b>Room / Location</b>   | <b>Room 610 (Virtual)</b>   | <b>Session Chair(s)</b> | <b>Dr. Tahesin Samira Delwar<br/>Dr. Md Aminul Islam</b> |
| <b>Session Moderator</b> | <b>Tanvir Hassan Ananta</b> | <b>Volunteer(s)</b>     | <b>Mehedi Hassan, Ifty Anwar</b>                         |

**Joining Information (Zoom):**

<https://bdren.zoom.us/j/98721132299?pwd=1LW8bUoxvNRtcJZ6127bzfBZ3xRgp6.1>

| # | Paper ID | Paper Title                                                                                                         | Author(s)                                                                                                                             |
|---|----------|---------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| 1 | 196      | Triple-Tone Assisted Multicarrier 5G Communication over Long-Distance OWC Links in Challenging Weather Environments | Muhammad Rahman*<br>Sohana Iqbal Tisha Tisha<br>Niloy R. Nath<br>Abdullah Al Musabbir<br>M. K. Juthi<br>M. Al Mamun<br>Shahriar Arfin |
| 2 | 271      | SHARP: Content-Fingerprint Anchoring for Self-Healing File Reference Resolution Across Local and Cloud Storage      | Arnob Aich*<br>Shamiul Islam<br>Md.Nasif Rafidi<br>Md. Sadman Hossain<br>Trisha Sarker<br>Samiha Islam Abrita                         |
| 3 | 301      | Impact of Metric Staleness on Kubernetes Horizontal Pod Autoscaling in Microservice Systems                         | Nishal Gurung*<br>Sushil Shrestha<br>Rajani Chulyadyo                                                                                 |

| # | Paper ID | Paper Title                                                                                      | Author(s)                                                                                                        |
|---|----------|--------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| 4 | 537      | Adaptive Financial Fraud Detection Using Heterogeneous Dynamic Graphs and Reinforcement Learning | Latul Hasan<br>Md Rashel Ahmmed<br>Md Didear Hossen<br>Muhim Bepari<br>Md. Faysal Ahmed<br>Md Fokrul Islam Khan* |
| 5 | 577      | Stochastic Modeling of Shoe Upper Defects in Footwear Manufacturing Using Monte Carlo Simulation | Md Sarfuddin Anowar Khan*<br>Syed Tahmid Jamil<br>Md Ariful Ferdous<br>Md. Manjumul Huque Mohin                  |
| 6 | 722      | A Physics Informed Machine Learning Approach For Predicting Solidification Cracking of Ti-4Al-6V | Md Shohanur Rahman*<br>Md. HAabibullah                                                                           |

#### VS8 | 26th Sep SAT 9:00 AM -10:30 AM

TRACK : IoT-IES(Internet of Things and Intelligent Embedded Systems)

|                   |                        |                  |                                                       |
|-------------------|------------------------|------------------|-------------------------------------------------------|
| Room / Location   | Room 610 (Virtual)     | Session Chair(s) | Dr. Engr. Mohammad Aman Ullah<br>Dr. Md. Maniruzzaman |
| Session Moderator | Ms. Joynob Binte Ahmed | Volunteer(s)     | Sayed Ibne Saif<br>Syed Sorfuddin Alif                |

#### Joining Information (Zoom):

<https://bdren.zoom.us/j/91519571555?pwd=jjsQ81pa8qNCjXBnaxbZlUeDIGq3vq.1>

| # | Paper ID | Paper Title                                                                                                            | Author(s)                                                                                                               |
|---|----------|------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| 1 | 281      | IoT-Enabled Multimodal Edge AI Baby Presence and Safety Monitoring System for Vehicles                                 | Anite Halim Sagor*<br>Dr. Md. Sheikh Sadi<br>Abdullah Al Galib Tonmoy                                                   |
| 2 | 283      | Edge-Enabled Deep Learning Framework for Predictive Maintenance in Industrial IoT Environments                         | Abdullah Al Galib Tonmoy*<br>MD SEYAM ALI<br>Md Imtiaz Ahmed<br>SHAH MD RIAZUL ISLAM Riaz<br>Debdatta Mallick           |
| 3 | 311      | Trust-Aware Lightweight Blockchain-Assisted Federated Learning Framework for Secure Resource-Constrained IoT Networks  | Abdullah Al Mamun Tasin<br>Md. Mujahid Hasnat<br>Sayadatul Arefin Kasherah<br>Abdullah Al Galib Tonmoy*<br>Roman Raihan |
| 4 | 371      | Design and Hardware Implementation of a 100kHz SEPIC Converter Using Microcontroller-based Fast PWM                    | Abdullah Atif*<br>Md. Nazmus Sakib<br>Abir Ahmed Hasnain                                                                |
| 5 | 379      | Power-Delay Trade-Off Analysis of Full Adder Designs Using CMOS, Transmission Gate, and ECRL Logic in 90 nm Technology | Fahmid Zawad*                                                                                                           |
| 6 | 640      | Real-Time Autonomous Lane Management on Embedded Hardware Using Multi-Camera Vision and YOLOv8                         | S.M. Shafayet Jamil<br>Md. Rakibul Hasan Nishat<br>Kazi Atif<br>Md. Meraj Ahmed*                                        |

#### VS9 | 26th Sep SAT 9:00 AM -10:30 AM

TRACK : ES-RCA(Electronic Systems, Robotics and Control Applications)

|                   |                     |                  |                                                 |
|-------------------|---------------------|------------------|-------------------------------------------------|
| Room / Location   | Room 610 (Virtual)  | Session Chair(s) | Dr. Shahid Md. Asif Iqbal<br>Dr. Nursadul Mamun |
| Session Moderator | Rizma Samanja Niloy | Volunteer(s)     | Tasnia Chowdhury, Chinmoy Barua Nobo            |

**Joining Information (Zoom):**

<https://bdren.zoom.us/j/95853395239?pwd=SeRLfXhyerDfVkedn9x5nRkf6jmuVo.1>

| # | Paper ID | Paper Title                                                                                                                    | Author(s)                                                                                                                            |
|---|----------|--------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| 1 | 245      | Adversarial Over-Sizing Detection in Campus Microgrids via Sinkhorn DRO                                                        | Sheikh Rahman*<br>Nadim Ahmed<br>Md Mohaiminul Islam Mahin<br>Lian Mollick Nehal<br>Md Fazle Hasan Shiblee                           |
| 2 | 251      | Temperature and State-of-Charge Aware Fast Charging for Lithium-Ion Battery Systems                                            | salman Himel<br>Fahmida Riya*<br>Shah Mohazzem Hossain                                                                               |
| 3 | 524      | Automated Insulator Fault Detection in High-Voltage Transmission Lines Using YOLOv12 and Aerial Imagery                        | Md. Absar*<br>Tanvir Mohammad Afnan<br>Simanta Chowdhury<br>Robiul Hossain Jeshan<br>Hridoy Chowdhury                                |
| 4 | 621      | Design and Performance Analysis of a Wideband THz MIMO Patch Antenna Using a Central Copper Barrier for Enhanced Isolation     | MD. RAKIB UDDIN*<br>Md. Ariful Islam<br>Md. Mehedi Hasan Tusar<br>Md. Mahmudul hasan Meem<br>A M F R S BIN K AYUB                    |
| 5 | 737      | A Unified MOPSO-Optimized Cascade Control Framework for DC–DC Boost Converters: Comparative Study of PI, PID, SMC, and FCS-MPC | Md Maniruz Zaman*<br>Jiad Bin Asad<br>Amit Das<br>Md. Shamsul Islam<br>Md. Mehedi Hasan Tusar<br>Uttam Mahapatra                     |
| 6 | 826      | An IoT-Compatible Machine Learning Framework for Sleep Apnea Risk Prediction Using SpO <sub>2</sub> and Heart Rate             | Md Muhasin Ali*<br>Md. Rakibul Islam<br>Abdullah Al Masud<br>Md Sadin Foraji<br>Saharia Hassan Safin<br>Israt Jahan Mojumdar         |
| 7 | 163      | An Improved Active Disturbance Rejection Control Framework for Robust Stabilization of a Two-Wheeled Self-Balancing Robot      | Touhedul Islam Tuhin*<br>Md. Nazmul Howlader<br>Asaduzzaman Rhine<br>Md. Masrur Ahmed Ahad<br>Safiullah Khan<br>Rimjhim Akter Bristi |